# Lecture 3

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# Lecture 3

1. Reading Summary – Chapter 3
2. Reading Summary – Chapter 4
3. Measures of Location (more mathematical notation than the text)
4. Measures of Spread (more mathematical notation than the text)
5. Vectors in R
6. Summary Statistics in R

Quiz 1 is today. Please remember to write your name on the quiz.
Quiz 2 is next Thursday in lab.

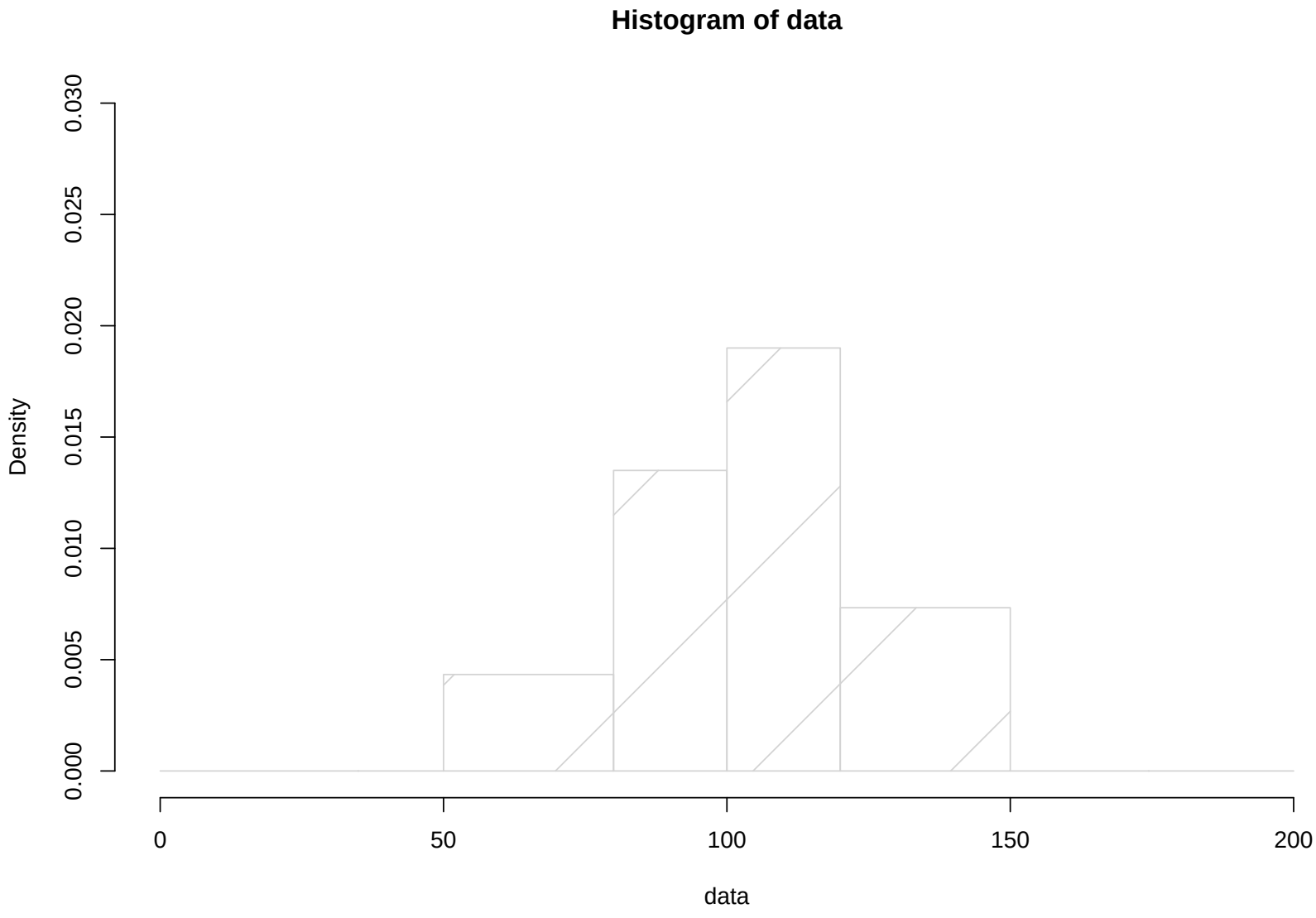
Read through Chapter 5 for the next lecture.

# Reading Summary – Chapter 3

# Section 3.1: Introduction

- A histogram is a graphical way to summarize a list of numbers.
- We divide the data into **class intervals**.
- Area represents percentages.
- No vertical scale is necessary.

# Discussion Question: What Percentage of Values are between 100 and 120?



# Section 3.2: Drawing a Histogram

- First create a **distribution table**.
  - Calculate the percentage of the data within each class interval.
  - You need to decide on an **endpoint convention** before you do this.
- Next calculate the height of the blocks.
  - The height is the percentage divided by the width of the interval.
- Then draw the histogram.
- Be sure to label the axes.

# Create a histogram

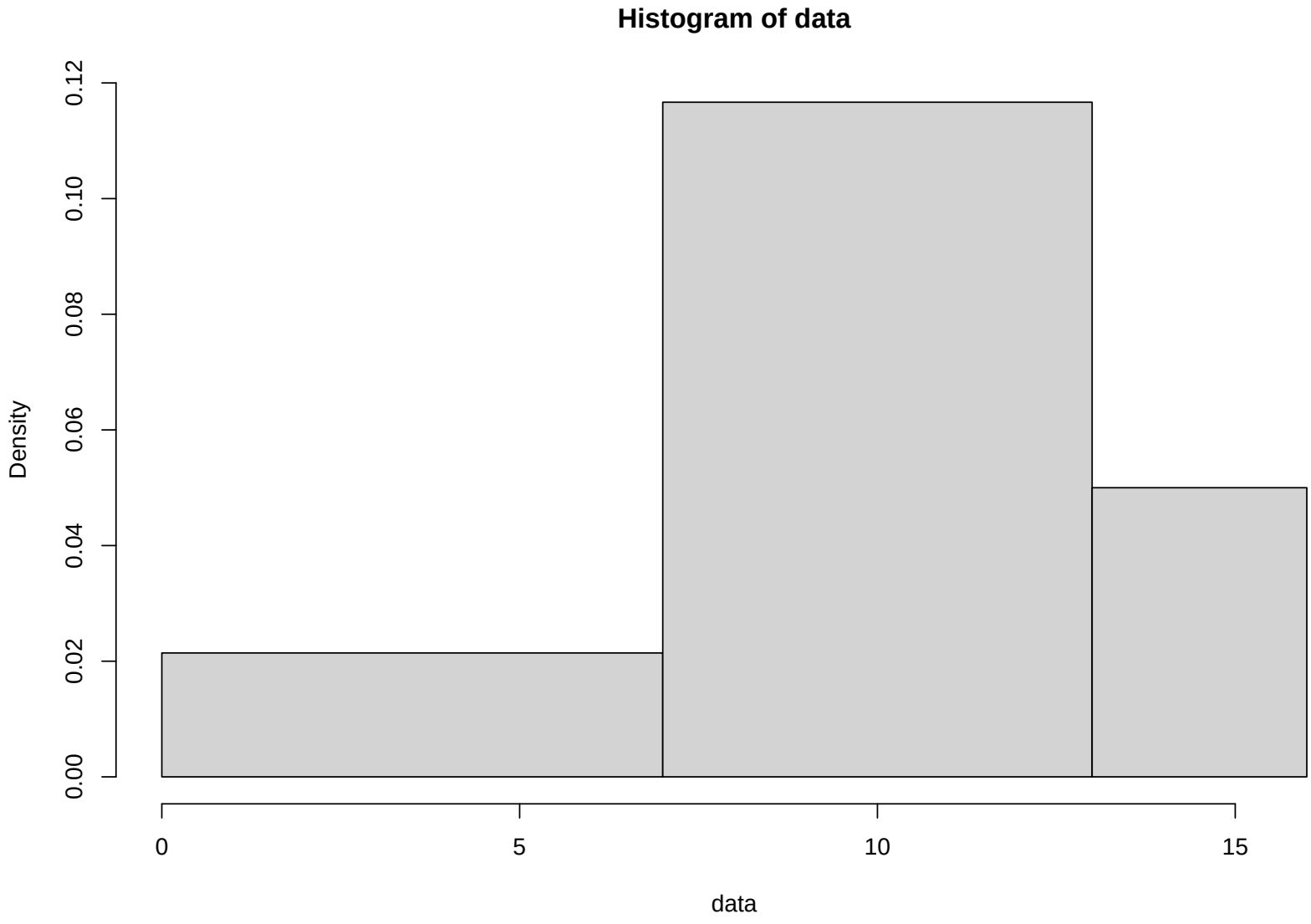
- Create a histogram for the following data:

```
##  [1] 10  6 12  5  8  9 14  8 11  6
```

```
##  [1] 10 11  9 10 10 11 10  7 15 15
```

| Interval | Count | Bar Height |
|---|---|---|
| 0 to $<$ 7 | 3 | (3/20)/(7 - 0) |
| 7 to $<$ 13 | 14 | (14/20)/(13 - 7) |
| 13 to $<$ 16 | 3 | (3/20)/(16 - 13) |

# The histogram



# Section 3.3: The Density Scale

- The vertical axis on a histogram is the **density scale.**
- The density scale measures "crowding" along the x-axis.
- The units of the density scale are "percent per X" where X is the units of the x-axis.
- Using the density scale is why area represents percentage.

# Section 3.4: Variables

- Variables can be **qualitative (categorical)** or **quantitative**.
- Quantitative variables can be **discrete** or **continuous**.
- When making a histogram of discrete variables, center the bars over the values.

# Section 3.5: Controlling for a Variable

- You can control for a variable in a observational study by making refined comparisons of histograms.
- Shifts in the histogram represent differences in the groups.
- The idea is basically the same as cross tabulation.

# Section 3.6: Cross Tabulation

- Introduction to cross tabulation.

# Section 3.7: Selective Breeding

- Interesting story; makes use of histograms.
  - Successive generations of rats were bred to be good (brights) or bad (dulls) at navigating mazes.
  - The brights were good at navigating mazes.
  - They weren't good at other cognitive tasks, however, and tended to be unsocial introverts.
  - The dulls, on the other hand, were bad at navigating mazes but tended to be more social.
  - This argued against a notion of generalized intelligence, at least in rats.

# Reading Summary – Chapter 4

# Section 4.1: Introduction

- Measures of location
- Measures of spread

# Section 4.2: The Average

- Introduction to the average (mean)
- Introduction to **cross sectional** and **longitudinal** studies.
  - A cross sectional study compares different groups of people at a specific point in time.
  - A longitudinal study follows people over time and considers how the outcome changes with time.

# Section 4.3: The Average and the Histogram

- The mean (average) is the point where the histogram balances.
- The median is the point where half the area of the histogram is on the left, and half on the right.
- If a histogram is symmetric, the median equals the mean.
- If a histogram has a long right tail, the mean is greater than the median.
- If a histogram has a long left tail, the mean is less than the median.

# Section 4.4: The Root-Mean-Square

- The root-mean-square (r.m.s.) of a list of numbers is the square root of the average of the numbers squared.

# Section 4.5: The Standard Deviation

- The standard deviation (SD) tells us how far away numbers on a list are from their average.
- Most of the numbers will be somewhere between 0 and 2 SDs away from the average.
- Rules of Thumb:
  1. About **two thirds** of the data will be within 1 SD of the average.
  2. About 95% of the data will be within 2 SDs of the average.

# Section 4.6: Computing the Standard Deviation

- For each number in a list, you can compute the deviation from average.
- Deviation from average = number - average.
- The standard deviation is the r.m.s. of the deviations from average.

# Section 4.7: Using a Statistical Calculator

- Many statistical calculators, and also the R programming language, don't calculate the $\mathrm{SD}$.
  - They actually calculate $\mathrm{SD}^+$.
- If $n$ is the number of numbers in the list, then $\mathrm{SD}^+ = \sqrt{\frac{n}{n-1}}\mathrm{SD}$.
- This amounts to dividing the sum of the squares of the differences from the mean by $n-1$ instead of $n$.
- $\mathrm{SD}^+$ is an **unbiased** estimate of the standard deviation.
- $\mathrm{SD}$ underestimates the standard deviation.
- The difference is only meaningful when $n$ is small.

# Measures of Location

- Suppose we have a list of $n$ numbers:

$$x_1, x_2, x_3, \ldots, x_n.$$

- What are some ways we can "summarize" this list with just a single number?
  - Mean
  - Median
  - Mode

# Mean

- The mean is just the ordinary average.
- We represent the mean with a bar over the variable.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

# Mean Example:

Consider the list:

$$3, 8, 4, 9, 1, 1, 2.$$

$$\bar{x} = \frac{1}{7}(3 + 8 + 4 + 9 + 1 + 1 + 2) = 4$$

# Median

- The median is the “middle” number in the list.
- First write the list in order, then take the middle number.

# Median Example:

Consider the list:

$$3, 8, 4, 9, 1, 1, 2.$$

First put the list in order:

$$1, 1, 2, 3, 4, 8, 9.$$

The median is the middle number: 3.

# Median – Technical Definition:

- Minor technical point: What to do if there are an even number of entries in the list?
- One possibility: Use the average of the two middle numbers.
- However, we will not use this convention.
- Below, we will give a precise definition of *percentile*.
- We will define the median as the "50-th percentile".
- In practice, this means the median is the smaller of the middle two numbers.

# Percentile – General Idea

- Basic idea: The $X$-th percentile is the number $q$ such that $X$ percent of the data is less than or equal to $q$.
- Example: Say a class takes a midterm. The $90^{th}$ percentile of the midterm scores would be the score where 90% of the class scored lower, and 10% percent higher.
- The exact definition of a percentile is tricky.
  - What if there are only 7 people in the class? There is no test score so that exactly 90% percent get lower, and 10% percent get higher.
  - Another example: What is the 90th percentile of

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10?$$

  - We could choose any number between 9 and 10.

# Percentile – Exact Definition

- The $X$th percentile of a list is the smallest number $q$ in the list such that at least $X$ percent of the numbers are less than or equal to $q$.
- Example: Consider the list

$$3, 8, 4, 9, 1, 1, 2.$$

What is the $75^{th}$ percentile?

# Percentile – Exact Definition 2

1. Put the numbers in order:

$$1, 1, 2, 3, 4, 8, 9.$$

2. Each number is $1/7 = 14.23$ percent of the dataset.
   - 28.57% of the numbers are less than or equal to 1.
   - 42.86% of the numbers are less than or equal to 2.
   - 57.14% are less than or equal to 3.
   - 71.43% are less than or equal to 4.
   - 85.71% are less than or equal to 8.
   - 100% are less than or equal to 9.

- The $75^{th}$ percentile is 8.

# Discussion Question – Percentiles

Consider the list:

$$3, 1, 4, 1, 5.$$

1. What is the $100^{th}$ percentile?
2. What is the $99^{th}$ percentile?
3. What is the $81^{st}$ percentile?
4. What is the $80^{th}$ percentile?
5. What about the $41^{st}$, $40^{th}$, $39^{th}$, $21^{st}$, $20^{th}$, and $19^{th}$?

Remember: The $X$th percentile of a list is the smallest number $q$ in the list such that at least $X$ percent of the numbers are less than or equal to $q$.

# Mode

- The most common number in the list.
- There can be more than one. Example: $3, 5, 2, 5, 3$.
- Researchers rarely report the mode.

# Mean or Median?

- Consider the list
$$1, 2, 3, 4, 1000.$$
- The mean is 202.
- The median is 3.
- Which number is better?
  - **IT DEPENDS ON CONTEXT!**
- The median is relatively unaffected by extreme values (outliers).
- The real question is, do we *care* about the outliers?

# Mean or Median – Examples

- Employee wages at a start-up
- House prices in a city
- Hourly data usage on your cell phone plan

# Mathematical Comment 1

Suppose we have a list of $n$ numbers:

$$x_1, x_2, x_3, \ldots, x_n.$$

Consider the quantity

$$\sum_{i=1}^{n} (x_i - a)^2.$$

This quantity is a minimum when $a$ is equal to the mean.

**The mean minimizes the total squared deviation.**

# Mathematical Comment 2

Consider the quantity

$$\sum_{i=1}^{n} |x_i - a|.$$

This quantity is a minimum when $a$ is equal to the median.

**The median minimizes the total absolute deviation.**

# Measures of Spread

- Standard Deviation
- Variance
- Interquartile Range (IQR)
- Median absolute deviation (MAD)

# Standard Deviation

$$\mathrm{SD} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

# Variance

The quantity

$$\mathrm{SD}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

is known as the **variance**.

The variance is theoretically useful, but is not a good measure of spread.

# Shortcut Formula for Variance

Note that:

$$\mathrm{Var}(x) = \overline{x^2} - \bar{x}^2.$$

The variance is the average of the squares minus the square of the average.

This formula is a short-cut to calculate the variance, and thus the standard deviation.

# Discussion Question - Shortcut Formula

Consider the list: $1, 2, 3$

Use the shortcut formula to calculate the standard deviation.

# Interquartile Range (IQR)

- The IQR is the $75^{th}$ percentile minus the $25^{th}$ percentile.
- Despite the name, this is a single number.

# Discussion Question - IQR

- What is the IQR of $3, 2, 5, 1, 3$?

# Median Absolute Deviation (MAD)

- The median of absolute deviations from the median.

$$\mathrm{MAD} = \mathrm{median}(|x_i - \mathrm{median}(x)|).$$

# Discussion Question - MAD

- What is the MAD of $3, 2, 5, 1, 3$?

# SD, IQR, or MAD?

- The decision of whether to use the SD, IQR, or MAD is very similar to the decision of whether to use the mean or median.
- The IQR and MAD are both relatively insensitive to outliers.

# Vectors in R

- A vector is just a list of numbers.
- In R, you can store the entire vector in one variable.
- We will create vectors in 4 main ways in Stat 280.
- The first is to simply list the numbers manually, using the c command.

```
a = c(1,2,5.7,9.4,0)
print(a)
```

```
## [1] 1.0 2.0 5.7 9.4 0.0
```

# Vectors in R (continued)

- The second way is to create a vector filled with a single number repeated many times, using the `rep` command.

```
a = rep(3.2, 6)
print(a)
```

```
## [1] 3.2 3.2 3.2 3.2 3.2 3.2
```

- The third way is to create a sequence of integers using a colon.

```
a = 5:12
print(a)
```

```
## [1]  5  6  7  8  9 10 11 12
```

# Columns in Data Frames as Vectors

- A **data frame** is how R represents a dataset.
- A data frame is a table.
  - The rows represent observations (people).
  - The columns represent variables (attributes) describing the observations.
- Each column in a data frame is a vector
- Use a dollar sign ($) to represent a column in data frame.

```
sample_data = data.frame(name=c("X","Y","Z"),
                         age=c(10,23,21),
                         year=c(1,3,2))
print(sample_data$age) #The age column of sample_data
```

```
## [1] 10 23 21
```

# Indexing

- Once we have a vector of numbers, we can refer to a specific entry in the vector using square brackets.
- For example, to refer to the third entry of the vector a, we would type a[3].

```
a = 5:12
print(a)
```

```
## [1]  5  6  7  8  9 10 11 12
```

```
print(a[3])
```

```
## [1] 7
```

# Summary Statistics in R

- In R, it is very easy to calculate:
  - mean,
  - median,
  - standard deviation, and
  - IQR.
- The IQR function uses a slightly different definition of IQR.
  - The IQR is still the difference between the $75^{th}$ and $25^{th}$ percentiles, but R calculates the percentiles differently.
  - See IQR calculation in R link for more details.

# R Output

```
x = c(3, 8, 4, 9, 1, 1, 2)
print(mean(x))
```

```
## [1] 4
```

```
#length() - number of numbers in the vector
n=length(x)
#The length, median, SD+, and IQR
print(c(n, median(x),sd(x),IQR(x)))
```

```
## [1] 7.000000 3.000000 3.265986 4.500000
```

```
print(sqrt((n-1)/n)*sd(x)) #The SD
```

```
## [1] 3.023716
```

# Review Questions

- See Review_questions3.pdf on Canvas